

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Branch: CSM

Section: B

Task No.: 1

Student Name: Appala Reddy

Roll No.: A24126552089

1. TITLE

Design and Develop a College Information Portal Using HTML and HTML5 to Demonstrate Semantic Structure, Text Formatting, Lists, Tables, Forms, and Multimedia Elements.

2. AIM

To design and develop a static HTML5 web page — the “College Information Portal” — that demonstrates the use of semantic tags, text formatting elements, ordered and unordered lists, hyperlinks, a structured table, images, an interactive registration form, HTML5 input types, and embedded audio/video elements.

3. OBJECTIVE

- To understand the basic structure and syntax of an HTML5 document.
- To use semantic tags such as header, nav, section, article, and footer to organise page content.
- To apply text formatting tags such as b, i, u, sup, and sub.
- To create ordered and unordered lists to present information.
- To insert hyperlinks that open external websites in a new tab.
- To design a table to display structured student data.
- To embed images with meaningful alt text.
- To build an interactive registration form using text, email, password, radio, checkbox, select, and textarea inputs.
- To use HTML5 input types such as date, number, color, and range.
- To embed audio and video elements using native HTML5 support.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create the structure of web pages. HTML5 introduces semantic elements and new form input types that make pages more meaningful and interactive, without needing external plugins or JavaScript.

In this page, semantic and formatting tags are used to structure content and present it clearly to the browser.

- header, nav, section, article, footer are used to organise related content into logical blocks.
- sup and sub raise or lower text for ordinal suffixes and small notes.
- ol and ul create ordered and unordered lists respectively.
- a href="..." target="_blank" creates a hyperlink that opens in a new tab.
- table, tr, th, td define a table, its rows, header cells, and data cells.
- form, input, select, textarea create interactive elements for user input.
- audio and video embed native media players without external plugins.

Basic flow of the page: Browser requests the HTML file → Browser parses the HTML → Elements are rendered as a structured web page

5. ALGORITHM / PROCEDURE

1. Create a new HTML file named college_portal.html.
2. Add the DOCTYPE declaration and set the language attribute.
3. Add the head section with meta tags and a page title.

4. Add a header section containing the college name, followed by a navigation bar with Home, About, and Contact links.
5. Create a section with formatted text, an ordered list of courses, and an unordered list of facilities.
6. Insert hyperlinks to external websites, set to open in a new tab.
7. Create a table to display student roll number, name, branch, and email.
8. Insert images for the college logo and campus.
9. Build a registration form using text, email, password, radio, checkbox, select, and textarea inputs.
10. Add an article block showing a notice.
11. Embed audio and video elements using the audio and video tags.
12. Add HTML5 input types such as date, number, color, and range.
13. Add a footer with copyright information.
14. Save the file and open it in a web browser.
15. Verify that all elements render and function correctly.

6. PROGRAM

college_portal.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>College Information Portal</title>
</head>
<body>

<header>
<h1>ANITS Engineering College</h1>
</header>

<nav>
<a href="#">Home</a> | <a href="#">About</a> | <a href="#">Contact</a>
</nav>

<section>
<p>Welcome to our college portal. We offer <b>quality education</b>,
<i>modern infrastructure</i>, and <u>excellent placements</u>.</p>

<p>Established in 2005 | Our college secured 1<sup>st</sup> rank
in the state for placements<sub> (2024)</sub>.</p>

<h2>Courses Offered</h2>
<ol>
<li>CSE</li>
<li>ECE</li>
<li>MECH</li>
</ol>

<h2>Facilities</h2>
<ul>
<li>Library</li>
<li>Sports Complex</li>
<li>Hostel</li>
</ul>

<p><a href="https://www.college-example.edu" target="_blank">College Website</a> |
<a href="https://www.google.com" target="_blank">Google</a></p>
</section>

<section>
<h2>Student Details</h2>
<table border="1" cellpadding="8">
<tr><th>Roll No</th><th>Name</th><th>Branch</th><th>Email</th></tr>
<tr><td>A24126552089</td><td>Appala Reddy</td><td>Computer
Science</td><td>koyyaappalareddy.24.csm@anits.edu.in</td></tr>
<tr><td>A24126552106</td><td>Maria Garcia</td><td>Electronics</td><td>maria.24.cam@.com</td></tr>
</table>

<h2>Logo & Campus</h2>


</section>

<section>
<h2>Student Registration Form</h2>
<form>
Name: <input type="text" required><br>
Email: <input type="email" required><br>
Password: <input type="password" required><br>
Gender: <input type="radio" name="gender">Male
<input type="radio" name="gender">Female
<input type="radio" name="gender">Other<br>
Hobbies: <input type="checkbox">Sports
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```
<input type="checkbox">Music
<input type="checkbox">Coding<br>
Branch:
<select>
<option>Computer Science</option>
<option>Electronics</option>
<option>Mechanical</option>
</select><br>
Comments:<br>
<textarea rows="4" cols="30"></textarea><br>
<input type="submit"><input type="reset">
</form>
</section>

<section>
<article>
<h3>Latest Notice</h3>
<p>Internal assessment activities will begin from next week.</p>
</article>

<h2>Media</h2>
<audio controls src="welcome.mp3"></audio>
<video controls width="300" src="campus_tour.mp4"></video>

<h2>HTML5 Input Types</h2>
<form>
Email: <input type="email"><br>
DOB: <input type="date"><br>
Age: <input type="number" min="17" max="30"><br>
Color: <input type="color"><br>
Rating: <input type="range" min="1" max="5">
</form>
</section>

<footer>
<p>&copy; 2026 ANITS Engineering College</p>
</footer>

</body>
</html>
```

7. CODE EXPLANATION

Tag / Attribute	Description
<header>, <nav>, <footer>	Group the banner, navigation links, and closing section into semantic, meaningful blocks.
<section>, <article>	Group related content and mark out a standalone block, respectively.
<sup>, <sub>	Raise or lower text for ordinal suffixes and small notes.
, <i>, <u>	Bold, italicise, and underline text for visual emphasis.
, ,	Create ordered and unordered lists and their items.
	Creates a hyperlink that opens in a new browser tab.
<table>, <tr>, <th>, <td>	Define a table, its rows, header cells, and data cells.
	Embeds an image with alternate text for accessibility.
<form>, <input>, <select>, <textarea>	Create the interactive elements of the registration form.
required	HTML5 built-in validation that blocks submission if a field is empty.
type="radio" / "checkbox"	Radio buttons allow one choice per group; checkboxes allow multiple.
type="date" / "number" / "color" / "range"	HTML5 input types for specialised data entry.
<audio controls>, <video controls>	Embed native audio/video players without external plugins.

8. TEST CASES AND EXECUTION DATA

The page was opened in a browser and each interactive element was tested individually; the revised results are recorded below.

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Open college_portal.html in a browser	Page loads and displays the college portal	Portal displayed with revised student details	Pass
TC-02	Enter Appala Reddy in the Name field and submit	Name is accepted and form validation proceeds	Name accepted successfully	Pass
TC-03	Enter an invalid email format	Browser shows an email-format validation error	Email validation message displayed	Pass
TC-04	Select Sports, Music and Coding checkboxes	All selected hobbies remain checked	All three checkboxes selected	Pass
TC-05	Click 'College Website' link	Link opens in a new browser tab	New tab opened correctly	Pass
TC-06	Drag the Rating slider	Slider value updates as it is dragged	Rating value updated	Pass
TC-07	Open the DOB field	Browser shows a calendar date-picker	Calendar picker displayed	Pass
TC-08	Open the Color field	Browser shows a colour-picker swatch	Colour picker displayed	Pass

9. OUTPUT



Welcome to Anil Neerukonda Institute of Technology and Sciences

Empowering Innovation and Excellence since 2001

[Google Search](#) | [Official College Website](#)

About Our Campus

Our institution provides **world-class education** with a focus on *practical learning*. We are accredited with Grade A+ by NAAC.

Key Highlights (Unordered List)

- Global Exchange Programs
- 100% Placement Assistance

Campus Life & Student Directory

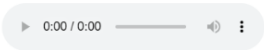


Student Details

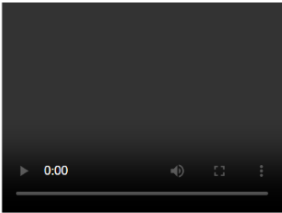
Roll Number	Name	Branch	Email
A24126552089	Koyya Appala Reddy	Computer Science	koyyaappalareddy.24.csm@anits.edu.in
A24126552106	Maria Garcia	Electronics	maria.24.cam@.com

Campus Media

College Anthem



Campus Tour Video



Student Registration Form

Full Name:

Email Address:

Password:

Date of Birth:

Age:

Gender:

☐ Male ☐ Female ☐ Other

Hobbies:

☐ Sports ☐ Music ☐ Coding

Select Branch:

Computer Science

Preferred ID Card Theme Color:

Blue

Self-Assessed Skill Level (1-10):

5

Comments / Statement of Purpose:

10. RESULT

Thus, a College Information Portal was successfully designed and developed using HTML5. Semantic tags, text formatting, ordered and unordered lists, hyperlinks, a structured table, images, a registration form, HTML5 input types, and multimedia elements were implemented and verified to render and function correctly in the web browser. The student details and execution data were updated for Appala Reddy.